

Vasculitis — Treatment: Induction & Maintenance



1. Induction banner

WHAT YOU SEE

An INDUCTION banner over the left half of the apothecary shop — the opening 'knock-down' phase pairing high-dose steroids with a second agent.

WHAT IT ENCODES

Induction therapy for severe/organ-threatening ANCA-associated vasculitis (RPGN, alveolar hemorrhage, severe neuropathy) = high-dose glucocorticoids PLUS a second agent — either cyclophosphamide or rituximab. Rituximab is non-inferior to cyclophosphamide (RAVE, RITUXVAS) and superior for relapsing PR3 disease. Non-severe disease (e.g., limited GPA) may use methotrexate or mycophenolate with steroids instead.

BOARD TRAP

Steroids ALONE are insufficient for severe AAV — choosing glucocorticoid monotherapy for RPGN or alveolar hemorrhage is the trap. You must add cyclophosphamide or rituximab for organ-threatening disease.

2. Cyclophosphamide warrior

WHAT YOU SEE

Muscular warrior labeled CYC (Cyclops)

WHAT IT ENCODES

Cyclophosphamide is an alkylating agent and a classic induction option for severe AAV: oral 2 mg/kg/day or IV pulse ~15 mg/kg every 2-3 weeks. IV pulse gives a lower cumulative dose with less leukopenia but slightly higher relapse than daily oral (CYCLOPS trial). Dose-reduce in renal impairment and the elderly; switch to a maintenance agent after ~3-6 months of induction.

BOARD TRAP

Cyclophosphamide is for INDUCTION only — continuing it long-term as maintenance is the trap because cumulative dose drives bladder cancer and infertility. Transition to rituximab or azathioprine once remission is achieved.

3. CYC toxicities

WHAT YOU SEE

Bladder cancer, infertility, AM dose + fluids tags near CYC

WHAT IT ENCODES

Cyclophosphamide toxicities: hemorrhagic cystitis and bladder cancer (from the metabolite acrolein — mitigate with MESNA + aggressive hydration and morning dosing so the drug isn't retained in the bladder overnight), gonadal failure/infertility (offer fertility preservation), myelosuppression, and infection (give PCP prophylaxis). Long-term bladder cancer surveillance is needed even after stopping.

BOARD TRAP

Forgetting MESNA/hydration and fertility counseling before cyclophosphamide is the trap. In young patients wishing to preserve fertility, rituximab is the better induction choice — defaulting to cyclophosphamide without considering this is a board error.

4. Rituximab — top rung

WHAT YOU SEE

RTX 500 mg x2 / q6mo, TOP RUNG (best) on relapse ladder

WHAT IT ENCODES

Rituximab (anti-CD20 B-cell depletion) is preferred induction for relapsing disease, PR3-ANCA disease, and patients wanting to preserve fertility; it is equally effective as cyclophosphamide for induction (RAVE) and SUPERIOR for maintenance/relapse prevention (MAINRITSAN), dosed roughly 500 mg every 6 months. Screen for hepatitis B (risk of reactivation) and give PCP prophylaxis.

BOARD TRAP

Prefer rituximab over cyclophosphamide in young patients desiring fertility and in relapsing disease — choosing cyclophosphamide there is the trap. Also screen HBV before rituximab: B-cell depletion can reactivate hepatitis B (a board safety point).

5. Prednisone moon

WHAT YOU SEE

Crescent moon labeled PRED, 1000 mg IV x3 pulse

WHAT IT ENCODES

Glucocorticoids are the backbone of induction across vasculitides: IV methylprednisolone pulses 1000 mg/day x3 for severe/organ-threatening disease, followed by oral prednisone ~1 mg/kg/day with a rapid taper. Modern protocols (PEXIVAS) favor REDUCED-dose steroid tapers — equally effective with fewer infections. Avacopan further enables steroid sparing.

BOARD TRAP

Prolonged high-dose steroids cause major toxicity (infection, osteoporosis, diabetes, AVN) — the modern trap is over-tapering too slowly/high. Reduced-dose regimens are non-inferior and safer; steroids alone are still inadequate for severe AAV (need a second agent).

6. MESNA / bladder protection

WHAT YOU SEE

MESNA / no bladder + infertility risk note near CYC arrow

WHAT IT ENCODES

MESNA (2-mercaptoethane sulfonate) binds acrolein, the toxic cyclophosphamide metabolite, in the bladder, and combined with aggressive hydration prevents hemorrhagic cystitis. Morning dosing of oral cyclophosphamide plus fluids avoids overnight bladder exposure. These measures reduce, but do not eliminate, long-term bladder cancer risk.

BOARD TRAP

MESNA protects the bladder but does NOT protect fertility or prevent bladder MALIGNANCY long-term — assuming MESNA makes cyclophosphamide 'safe' is the trap. Separate counseling on infertility and ongoing bladder cancer surveillance is still required.

7. Plasma exchange — limited

WHAT YOU SEE

PLASMA EXCHANGE only ANCA + severe, >48 hrs from RRT, dialysis-dependent

WHAT IT ENCODES

Plasma exchange (PLEX) has a LIMITED role in AAV: the PEXIVAS trial showed it does NOT improve death or ESRD overall. Reserve it for selected severe cases — rapidly rising creatinine/dialysis-dependent severe RPGN, possibly diffuse alveolar hemorrhage, and concomitant anti-GBM (double-positive) disease. PLEX is clearly indicated in anti-GBM/Goodpasture disease.

BOARD TRAP

Routine PLEX for ALL severe AAV is NOT recommended (PEXIVAS) — adding it reflexively for any severe ANCA vasculitis is the trap. It is, however, standard in anti-GBM disease and reasonable in selected dialysis-dependent RPGN/DAH.

8. Prophylaxis (PCP)

WHAT YOU SEE

PROPHYLAXIS, PCP prophylaxis, RTX maintenance tags lower left

WHAT IT ENCODES

Patients on induction immunosuppression (high-dose steroids + cyclophosphamide or rituximab) require *Pneumocystis jirovecii* (PCP) prophylaxis, typically TMP-SMX (one DS tab 3x/week or SS daily). TMP-SMX additionally reduces relapses in upper-airway GPA. Also update vaccinations and screen for latent infections (TB, HBV) before/during immunosuppression.

BOARD TRAP

Omitting PCP prophylaxis in patients on steroids + cyclophosphamide/rituximab is a classic safety trap — *Pneumocystis pneumonia* is a preventable, life-threatening complication. The risk is highest with high-dose steroids plus a steroid-sparing agent and lymphopenia.

9. Bisphosphonate

WHAT YOU SEE

BISPHOSPHONATE tag bottom left

WHAT IT ENCODES

Glucocorticoid-induced osteoporosis prophylaxis: calcium + vitamin D for everyone on chronic steroids, plus a bisphosphonate for those at moderate/high fracture risk or on prolonged high-dose steroids. Bone loss is fastest in the first 3-6 months, so start prevention EARLY rather than waiting for a low DXA.

BOARD TRAP

Waiting for an abnormal DXA before starting bone protection is the trap — initiate prophylaxis when committing a patient to prolonged steroids, because the fastest bone loss occurs early.

10. Avacopan — steroid sparing

WHAT YOU SEE

STERIOD SPARING, AVACOPAN, 7 remission @ 52 wk panel center

WHAT IT ENCODES

Avacopan is an oral C5a-RECEPTOR antagonist (blocks neutrophil activation via complement) approved as an adjunct in severe AAV. The ADVOCATE trial showed it enables marked glucocorticoid reduction with superior sustained remission at 52 weeks and less steroid toxicity. Monitor liver enzymes (hepatotoxicity reported).

BOARD TRAP

Avacopan is a steroid-SPARING add-on to standard induction (rituximab/cyclophosphamide), NOT a replacement for the induction immunosuppressant. Using avacopan + steroids alone without a B-cell-depleting/alkylating agent for severe AAV is the trap.

11. Maintenance ladder

WHAT YOU SEE

MAINTENANCE banner, RELAPSE RATE LADDER

WHAT IT ENCODES

Maintenance therapy (after remission) prevents relapse, ranked by efficacy: rituximab (best — lowest relapse, MAINRITSAN) > azathioprine ≈ methotrexate > mycophenolate mofetil (worst — highest relapse, IMPROVE trial). Continue maintenance for at least ~18-24 months, often longer in PR3/relapsing disease. Glucocorticoids are tapered to low dose or off.

BOARD TRAP

MMF is the LEAST effective maintenance agent in AAV (higher relapse than azathioprine) — selecting MMF as first-line maintenance is the trap. Rituximab is the preferred maintenance agent, especially for relapsing/PR3 disease.

12. AZA / MTX rungs

WHAT YOU SEE

AZA second rung, MTX third rung (TPMT, folate), MMF bottom rung

WHAT IT ENCODES

Oral maintenance options below rituximab: azathioprine (check TPMT activity first to avoid life-threatening myelosuppression; avoid with allopurinol) and methotrexate (weekly, with folate supplementation; monitor LFTs/CBC). Both are stepped down from cyclophosphamide induction once remission is achieved.

BOARD TRAP

Methotrexate is contraindicated in significant renal impairment (accumulation → toxicity) and in active severe disease — using MTX for a patient with reduced GFR or organ-threatening AAV is the trap. Likewise, starting azathioprine without checking TPMT (or alongside allopurinol) risks severe marrow suppression.

13. Caution circle — not for GPA

WHAT YOU SEE

CAUTION CIRCLE, NOT for GPA, CYC + RTX combo = NO benefit + ↑ infection

WHAT IT ENCODES

Combining cyclophosphamide AND rituximab for induction adds no proven benefit while increasing infection and toxicity — choose ONE induction agent. Standard severe-AAV induction is glucocorticoids + (rituximab OR cyclophosphamide), not both together.

BOARD TRAP

Stacking cyclophosphamide + rituximab to be 'more aggressive' is the trap — it raises infection risk without improving remission. Pick a single induction immunosuppressant alongside steroids (± avacopan).

14. Duration ~12-24 mo

WHAT YOU SEE

Clock ≥ 2 years near maintenance arrow

WHAT IT ENCODES

Maintenance immunosuppression continues for at least ~18-24 months after achieving remission, and longer (sometimes indefinitely) for PR3-ANCA or relapsing disease given high relapse rates. Stopping too early is a major cause of relapse. Periodically reassess organ function, drug toxicity, and ANCA/clinical status.

BOARD TRAP

Discontinuing maintenance at 6-12 months — especially in PR3/relapsing disease — is the trap, as relapse risk is high. Extend maintenance (often 2+ years) and taper cautiously based on serotype and relapse history.

15. ANCA rise = relapse risk

WHAT YOU SEE

ANCA RISE = RELAPSE risk, t for GCA/PMR lower right tags

WHAT IT ENCODES

A rising ANCA titer (especially PR3-ANCA) and reappearance of ANCA are associated with INCREASED relapse risk, but a titer change alone is not sufficient to diagnose relapse or restart therapy. Treat clinical/biopsy-proven relapse; use rising titers to heighten surveillance, not to drive blind escalation. (GCA/PMR are treated with glucocorticoids $\pm$ tocilizumab/methotrexate, a separate pathway from AAV.)

BOARD TRAP

Escalating immunosuppression on a rising ANCA alone — without clinical evidence of relapse — is the trap (risks over-treatment/infection). Increase monitoring instead and treat when clinical or histologic relapse appears.
